

7 The table shows some information about the polypeptides that make up haemoglobin.

	α -globin	β -globin
total number of amino acid residues in polypeptide chain	141	146
position of amino acid cysteine in polypeptide chain	104	93 and 112

Scientists studied the region of the β -globin polypeptide chain containing the amino acid cysteine at position 93. They found that:

- this region faces outwards when **no** oxygen is attached to the haem group
- this region faces inwards when oxygen is attached to the haem group
- replacing cysteine with a different amino acid reduces the Bohr shift.

What can be concluded from the information about cysteine in haemoglobin?

- A** More than 1% of the amino acids in one haemoglobin protein are cysteine.
- B** In α -globin, there is a cysteine closer to the end of the polypeptide chain with an unreacted carboxyl group than in β -globin.
- C** The replacement of the cysteine at position 93 in β -globin decreases the affinity of haemoglobin for oxygen at low pH.
- D** The binding of oxygen to the haem group causes the region of β -globin containing cysteine at position 93 to become more hydrophilic.